CUSTOMER SALES ANALYSIS

Week 5 Internship Project | Python, Pandas & Data Visualization

Prepared for The Developers Arena internship task. This report describes the project workflow and the metrics to calculate. Add verified findings after running the notebook on the supplied CSV files.

1. Project Overview

The objective is to analyze sales transactions, explore product and regional performance, identify monthly trends, and inspect customer churn data. Python and Pandas support loading, cleaning, transforming, aggregating, and summarizing data. Matplotlib and Seaborn create visualizations.

2. Datasets

Dataset	Visible columns	Purpose
sales_data.csv	Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales	Revenue, product demand, regional and monthly trends.
customer_churn.csv	CustomerID, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen, Churn	Explore churn counts and possible patterns by customer attributes.

Data note: The files may describe different customer populations. Verify the identifiers before merging; Customer_ID and CustomerID should not be assumed to identify the same people.

3. Methodology

• **Load and explore:** Read the CSV files and inspect sample rows, shape, columns, and data types.

• **Data quality:** Check missing values and duplicates. Investigate missing or invalid values before deciding how to handle them.

• **Transform:** Clean column-name whitespace, convert Date to datetime, and extract year, month, and day where dates are valid.

• **Aggregate and filter:** Calculate sales totals and averages; group by product or region; and filter using multiple conditions.

• **Summarize:** Create pivot tables for product and regional sales.

• **Visualize:** Create monthly sales, revenue-by-product, regional performance, product quantity, and churn-count charts.

4. Planned Visualizations

Chart	Purpose
Monthly Sales Performance	Show how recorded sales change over time.

Chart	Purpose
Revenue by Product	Compare total sales by product.
Regional Sales Performance	Compare sales across regions.
Product Demand Analysis	Compare total quantity sold by product.
Customer Churn Distribution	Count the records in each churn category.

5. Metrics to Record

• Total revenue: sum of Total_Sales.

• Average transaction sales: mean of Total_Sales.

• Top product and region: category with the highest grouped Total_Sales.

• Monthly trend: sum of Total_Sales by valid year-month.

• Churn distribution: count of each Churn value; verify the dataset documentation before interpreting values such as 0 and 1.

6. Findings and Recommendations

Complete this section after running the notebook. Replace the prompts below with results calculated from the supplied files:

• Finding 1: [Highest-revenue product and its value.]

• Finding 2: [Highest-revenue region and its value.]

• Finding 3: [Monthly or seasonal pattern supported by the data.]

• Finding 4: [Churn distribution and any supported pattern by contract.]

7. Limitations

No CSV files were available to calculate actual results while preparing this template, so no revenue totals, rankings, customer counts, or churn rates are invented. Product-pair analysis requires an order identifier and item-level records. Merge customer data only when the identifiers are confirmed to match.

8. Conclusion

The project demonstrates a practical workflow for customer and sales data analysis using Python. After executing the notebook, add the verified results and generated charts to the report before final submission.